MEEN 305 Final Project

Minimum-Weight Beam Design and Analysis
Tapered Hollow Rectangular Tube in PLA

Team #6

Spring 2026

Contents

1 Problem Statement and Requirements **2**
- 1.1 Fixed Project Inputs . . . 2
- 1.2 Objective . . . 2
- 1.3 Design Variables . . . 2

2 Material Selection **2**
- 2.1 Available FDM Polymers . . . 2
- 2.2 Orientation and Infill Factors . . . 3
- 2.3 Justification for PLA . . . 3

3 Beam Parametrization **3**
- 3.1 Cross-Section Topology . . . 3
- 3.2 Profile Functions . . . 4
- 3.3 Section Properties . . . 4

4 Governing Equations and Analysis Methodology **6**
- 4.1 Statics: Reactions and Internal Loads . . . 6
- 4.2 von Mises Stress . . . 6
- 4.3 Factor of Safety . . . 6
- 4.4 Deflection via Euler-Bernoulli Integration . . . 7
- 4.5 Beam Weight . . . 7
- 4.6 Assumption: Self-Weight Neglected . . . 7
- 4.7 Equation Dependency Chain . . . 7

5 Engineering Judgment Constraints **8**

6 Optimization Methodology **8**
- 6.1 Search Strategy . . . 8
- 6.2 Objective Function . . . 9
- 6.3 Why Load Case 2 Governs for Team 6 . . . 9
- 6.4 Printability Bounds . . . 9

7 Final Optimized Design **10**

7.1 Geometry . . . 10
7.2 Profile Equations . . . 10
7.3 Section Properties Profiles . . . 11

8 Load Case Analysis Results **11**
8.1 Summary Table . . . 11
8.2 Governing Load Case . . . 11
8.3 Load Case 1: Upright Bending . . . 11
8.4 Load Case 2: Side-Rotated Bending (Governing) . . . 12

9 Optimization History and Design Comparison **14**

10 Numerical Substitution Summary **14**
10.1 Effective Material Properties . . . 14
10.2 Load Case 1 Key Values . . . 14
10.3 Load Case 2 Key Values (Governing) . . . 14

11 Assumptions and Limitations **15**

12 Print Plan **15**

13 Symbol Table **16**

14 Conclusion **16**

1 Problem Statement and Requirements

1.1 Fixed Project Inputs

The beam is simply supported with the following prescribed boundary conditions and loading:

Parameter	Symbol	Value
Support span	L	8.0 in
Total beam length	L_{total}	9.0 in
Load Case 1 magnitude	P_1	5.0 lbf
Load Case 1 location	a_1	4.0 in from support A (midspan)
Load Case 1 orientation	—	Upright (load applied vertically)
Load Case 2 magnitude	P_2	8.0 lbf $\quad (2 + \text{Team\#} = 2 + 6)$
Load Case 2 location	a_2	5.0 in from support A $\quad (2 + \frac{6}{2} = 5)$
Load Case 2 orientation	—	Rotated 90° about span axis
Required factor of safety	FoS_{min}	≥ 1.5 for both load cases

1.2 Objective

Minimize the weight of the printed beam,

$$\text{minimize} \quad W = \rho \int_0^L A(x)\, dx,$$

subject to the factor-of-safety constraint for both load cases and all engineering and manufacturing constraints defined in Section 5.

1.3 Design Variables

The beam cross-section is a tapered hollow rectangular tube with spanwise lightening openings. The geometry is fully described by **10 continuous design variables**:

Variable	Symbol	Units	Role
Left/mid/right outer width	b_L, b_M, b_R	in	Tube width at $x = 0, L/2, L$
Left/mid/right outer height	h_L, h_M, h_R	in	Tube height at $x = 0, L/2, L$
Wall thickness	t	in	Uniform flange/web thickness
Left/mid/right opening ratio	r_L, r_M, r_R	–	Fraction of inner web height removed

All six profile dimensions (b, h, r) vary continuously along the span via piecewise linear interpolation between the three control points. Wall thickness t is kept spatially uniform, consistent with FDM slice practice.

2 Material Selection

2.1 Available FDM Polymers

The Rapid Prototyping Studio (JCAIN 422) stocks four filament options. Their nominal bulk properties and effective FDM properties are:

Material	E [psi]	σ_{allow} [psi]	ρ [lbf/in^3]	k_{orient}	k_{infill}	E_{eff} [psi]
PLA	500 000	4 000	0.045	0.70	0.85	297 500
ASA	380 000	3 600	0.043	0.65	0.85	209 950
PETG	290 000	3 500	0.046	0.72	0.85	177 480
TPU	10 000	2 500	0.043	0.75	0.90	6 750

2.2 Orientation and Infill Factors

FDM parts are mechanically anisotropic because deposited layers bond through partial re-melting. The interlayer bond is the dominant failure surface. Two reduction factors are applied multiplicatively to all nominal isotropic properties:

- **Orientation factor** k_{orient}: accounts for the reduction in strength and stiffness perpendicular to the print plane. When the span axis lies along the print direction (layer lines parallel to the neutral axis), load is transferred through the solid infill struts rather than across layer interfaces. We use $k_{orient} = 0.70$ for PLA, consistent with RPS guidance that layer lines are "the single biggest failure point."
- **Infill factor** k_{infill}: accounts for the reduced solid volume at less-than-100% rectilinear infill. We use $k_{infill} = 0.85$, corresponding to approximately 40% rectilinear infill with solid top/bottom shells.

Effective properties:

$$E_{eff} = k_{orient}\, k_{infill}\, E_{bulk} = 0.70 \times 0.85 \times 500\,000 = 297\,500 \text{ psi}$$

$$\sigma_{allow,eff} = k_{orient}\, k_{infill}\, \sigma_{allow} = 0.70 \times 0.85 \times 4\,000 = 2\,380 \text{ psi}$$

2.3 Justification for PLA

PLA is selected for the following engineering reasons:

1. **Highest effective stiffness**: $E_{eff,PLA} = 297\,500$ psi versus 209 950, 177 480, and 6 750 psi for ASA, PETG, and TPU respectively. Deflection is inversely proportional to $E_{eff}I$; PLA permits the smallest cross-section for a given deflection.
2. **Highest allowable stress**: $\sigma_{allow,eff,PLA} = 2\,380$ psi gives the largest FoS margin for a given cross-section, enabling more material removal.
3. **Printability**: PLA is the recommended default at the RPS and requires no special enclosure or bed adhesion, making reliable prints more likely.
4. **Density**: PLA density (0.045 lbf/in^3) is representative of structural polymers; ASA is marginally lighter but at the cost of significantly lower effective properties.

TPU is immediately eliminated (flexible elastomer, not a structural polymer). ASA and PETG are eliminated because both have lower E_{eff} and lower $\sigma_{allow,eff}$ than PLA — the same geometry would be heavier and weaker.

3 Beam Parametrization

3.1 Cross-Section Topology

At each spanwise location x, the cross-section is a hollow rectangular tube with a centered web opening (lightening slot) in both vertical web faces. The outer dimensions at position x are $b(x) \times$

$h(x)$; the wall thickness t is uniform throughout. The opening removes a fraction $r(x)$ of the clear inner web height $h_{clear}(x) = h(x) - 2t$ from both web faces, leaving solid flanges top and bottom.

3.2 Profile Functions

Each of the three continuous profiles is defined by piecewise linear interpolation between three control points at $x = 0$ (left support), $x = L/2 = 4$ in (midspan), and $x = L = 8$ in (right support):

$$b(x) = \begin{cases} b_L + \dfrac{2(b_M - b_L)}{L}\, x, & 0 \le x \le \frac{L}{2} \\ b_M + \dfrac{2(b_R - b_M)}{L}\,(x - \frac{L}{2}), & \frac{L}{2} < x \le L \end{cases}$$

and analogously for $h(x)$ and $r(x)$. This three-point piecewise-linear scheme provides enough shape freedom to represent arch profiles, linearly tapered beams, and uniform beams as special cases, while keeping the number of design variables tractable.

3.3 Section Properties

Cross-sectional area (with openings).

$$A(x) = 2\, b(x)\, t + [1 - r(x)]\, 2\, t\, h_{clear}(x), \quad h_{clear}(x) = h(x) - 2t$$

The first term is the two horizontal flanges; the second is the remaining vertical web area after the lightening slot is removed.

Second moment of area — upright bending (Load Case 1). The full hollow-tube second moment minus the two opening slots:

$$I_{upright}(x) = \frac{b(x)\, h(x)^3 - [b(x) - 2t]\, [h(x) - 2t]^3}{12} - 2\, \frac{t\, [r(x)\, (h(x) - 2t)]^3}{12}$$

Second moment of area — side-rotated bending (Load Case 2). The beam is physically rotated 90° about the span axis, so b becomes the vertical (bending) dimension and h becomes the lateral dimension:

$$I_{side}(x) = \frac{h(x)\, b(x)^3 - [h(x) - 2t]\, [b(x) - 2t]^3}{12} - 2\, \frac{t\, [r(x)\, (b(x) - 2t)]^3}{12}$$

Section modulus. The section modulus uses $c_y(x)$ for upright bending (extreme fiber in the y-direction, i.e. half the outer height) and $c_z(x)$ for side-rotated bending (extreme fiber in the z-direction, i.e. half the outer width):

$$S_{upright}(x) = \frac{I_{upright}(x)}{c_y(x)}, \qquad c_y(x) = \frac{h(x)}{2}$$

$$S_{side}(x) = \frac{I_{side}(x)}{c_z(x)}, \qquad c_z(x) = \frac{b(x)}{2}$$

Shear area.

$$A_{shear}(x) = [1 - r(x)]\, 2t\, h_{clear,active}(x)$$

where $h_{clear,active}$ is the clear inner dimension in the active bending orientation: $h_{clear}(x) = h(x) - 2t$ for upright, $(b(x) - 2t)$ for side.

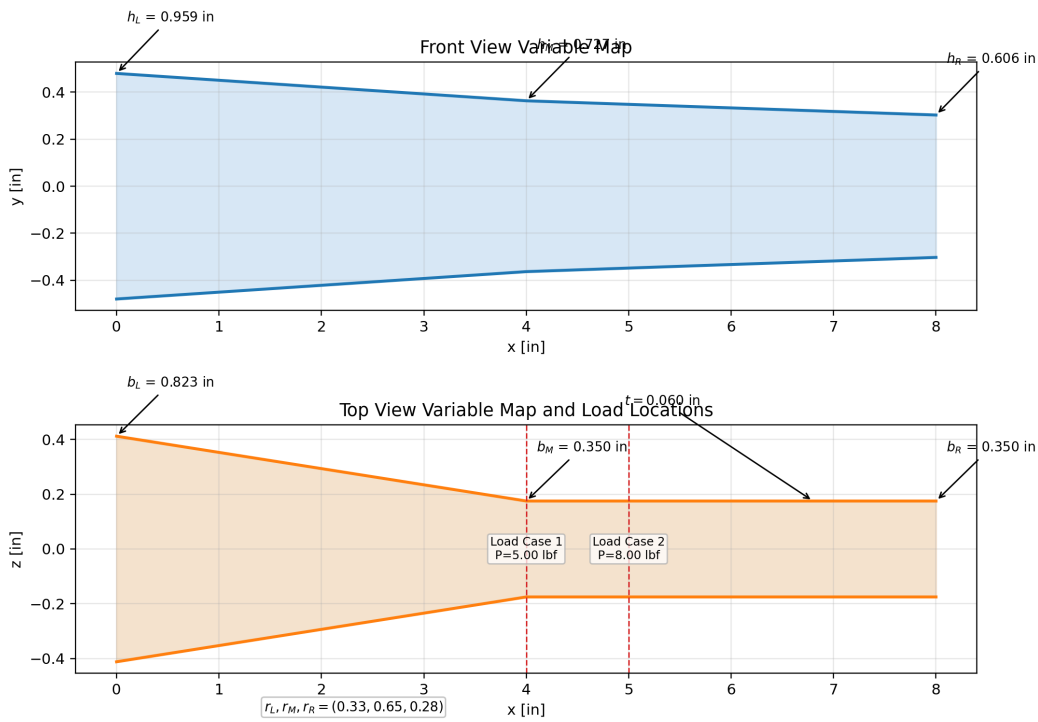

Figure 1: Variable map for the final optimized design. Front view (top) shows the height profile $h(x)$ with control-point annotations; top view (bottom) shows the width profile $b(x)$, wall thickness t, and load locations.

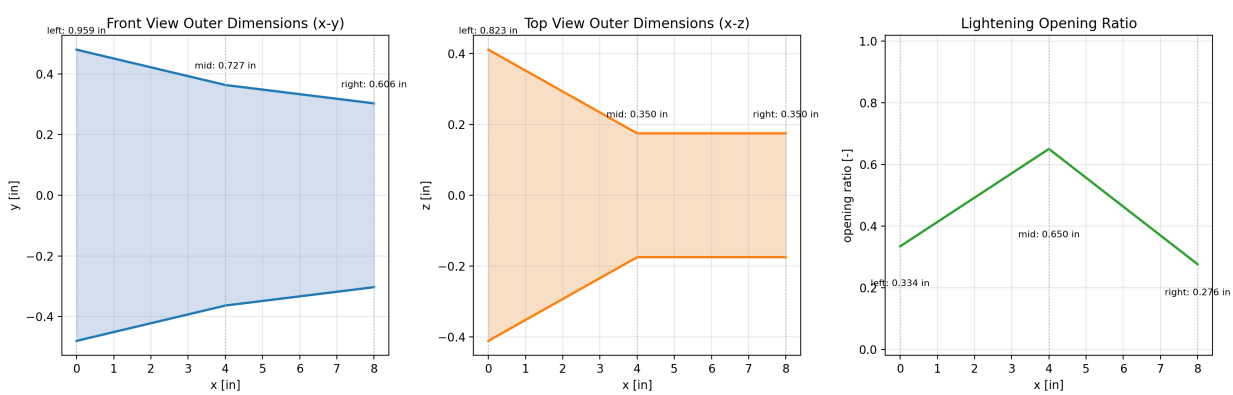

Figure 2: Outer dimension profiles and lightening-opening ratio for the final design. Left: front-view height profile $h(x)$; centre: top-view width profile $b(x)$; right: web opening ratio $r(x)$ along the span.

4 Governing Equations and Analysis Methodology

4.1 Statics: Reactions and Internal Loads

For a simply supported beam with one point load P at $x = a$:

$$R_A = \frac{P(L-a)}{L}, \qquad R_B = \frac{Pa}{L}$$

Shear force and bending moment (piecewise, with the load-point discontinuity):

$$V(x) = \begin{cases} R_A & 0 \le x < a \\ R_A - P & a < x \le L \end{cases} \qquad M(x) = \begin{cases} R_A\, x & 0 \le x < a \\ R_A\, x - P(x-a) & a < x \le L \end{cases}$$

4.2 von Mises Stress

The von Mises criterion combines all stress components into one scalar that can be compared directly against the effective allowable stress. The three intermediate quantities that feed into it are:

Normal bending stress. Using the Euler-Bernoulli flexure formula at the extreme fiber (distance c_y or c_z from the neutral axis, depending on orientation):

$$\sigma_b(x) = \frac{|M(x)|}{S(x)}$$

Shear stress. Shear is distributed over the active web area remaining after the lightening opening. A conservative uniform-distribution approximation is used:

$$\tau(x) = \frac{|V(x)|}{A_{shear}(x)}$$

von Mises equivalent stress. For the plane-stress state at the critical point (bending-dominant at the extreme fiber, shear-dominant at the neutral axis):

$$\sigma_{vm}(x) = \sqrt{\sigma_b(x)^2 + 3\,\tau(x)^2}$$

The σ_b^2 term dominates wherever $|M(x)|$ is large (e.g. midspan); the $3\tau^2$ term dominates near the supports where $M \to 0$ but V is nonzero. The governing (maximum) von Mises stress determines the minimum FoS over the span.

4.3 Factor of Safety

$$\mathrm{FoS}(x) = \frac{\sigma_{allow,eff}}{\sigma_{vm}(x)}$$

The governing (minimum) FoS is tracked at every discretized station x across both load cases:

$$\mathrm{FoS}_{min} = \min_{x,\,\mathrm{LC}} \mathrm{FoS}(x)$$

The design constraint is $\mathrm{FoS}_{min} \ge 1.5$.

4.4 Deflection via Euler-Bernoulli Integration

The Euler-Bernoulli beam equation for a non-uniform cross-section:

$$E_{eff}\, I(x)\, v''(x) = M(x)$$

Because $I(x)$ varies continuously with x, slope $\theta(x) = v'(x)$ and deflection $v(x)$ are obtained by numerical integration (trapezoidal rule, 801 stations) with simple-support boundary conditions $v(0) = v(L) = 0$:

$$\theta(x) = \int_0^x \frac{M(s)}{E_{eff}\, I(s)}\, ds + C_1$$
$$v(x) = \int_0^x \theta(s)\, ds + C_1\, x$$

The integration constant C_1 is determined by enforcing $v(L) = 0$.

4.5 Beam Weight

$$W = \rho \int_0^L A(x)\, dx$$

The integral is evaluated numerically over the same 801-station grid.

4.6 Assumption: Self-Weight Neglected

Self-weight is not included in the shear, moment, stress, or deflection calculations. Justification:

- The optimized beam weighs $W = 0.03454$ lbf.
- Applied point loads total $P_1 + P_2 = 5 + 8 = 13$ lbf.
- Self-weight is $0.03454/13 \approx 0.27\%$ of applied load.
- Standard practice neglects distributed self-weight when it is less than 5% of applied load. At 0.27%, the assumption is conservative by more than an order of magnitude; including self-weight would increase midspan stress by $< 0.3\%$.

4.7 Equation Dependency Chain

The full analysis pipeline from inputs to objective is:

$$(L,\, P,\, a) \xrightarrow{\text{statics}} R_A,\, R_B \;\rightarrow\; V(x),\, M(x) \xrightarrow{\text{geometry}} A(x),\, I(x),\, S(x) \;\rightarrow\; \sigma_b,\, \tau \;\rightarrow\; \sigma_{vm} \;\rightarrow\; \text{FoS}$$

$$M(x),\, E_{eff},\, I(x) \xrightarrow{\text{integration}} v(x) \qquad\qquad A(x),\, \rho \xrightarrow{\text{integration}} W$$

Both load cases use the same geometry functions; the active bending orientation switches between $I_{upright}$, c_y (LC1) and I_{side}, c_z (LC2).

5 Engineering Judgment Constraints

An idealized stress analysis would permit geometries that are physically unrealizable on an FDM printer. The following engineering judgment constraints are hard-coded in `beam_solver.py` (`TaperedRectangularT`

1. **Minimum cross-section at every station, including supports.** At a simply supported beam end, the bending moment is zero, so idealized theory requires zero section modulus — i.e., zero cross-section. In reality, the beam must physically contact its supports and transfer shear. The code enforces $h(x), b(x) \geq 0.15$ in (≈ 3.8 mm) at every station.

2. **Minimum printable wall thickness.** A purely stress-based analysis would permit paper-thin walls wherever stress is low. On a 0.4 mm nozzle, fewer than two continuous perimeters (< 0.8 mm) produces unreliable, porous walls. The code enforces $t \geq 0.040$ in (≈ 1.0 mm).

3. **Wall thickness must fit inside the cross-section.** $t < b/2$ and $t < h/2$ at every station; otherwise the tube wall would overlap the centerline.

4. **Ligament constraint for web openings.** The remaining solid ligament on each side of a web opening must exceed the wall thickness: $\frac{1}{2}(1-r)\, h_{clear} > t$. This ensures the web-to-flange junction is thick enough for load transfer and printing.

5. **Opening ratio bounded below 0.95.** A ratio of 1.0 would remove the web entirely, leaving two unconnected flanges. The code enforces $r < 0.95$.

The **optimizer search bounds** are more restrictive than these hard minimums, providing the primary engineering envelope:

Variable	Optimizer minimum	Optimizer maximum
Height h [in]	0.38	1.30
Width b [in]	0.35	1.60
Wall thickness t [in]	0.06	0.18
Opening ratio r [–]	0.00	0.87

The minimum height of 0.38 in (≈ 9.7 mm) ensures a realistic cross-section at every station including at the supports. The maximum opening ratio of 0.87 was chosen to guarantee at least 1.9 mm of solid ligament per side at the minimum-height station — about 5 nozzle widths at 0.4 mm, sufficient for a reliable continuous wall.

6 Optimization Methodology

6.1 Search Strategy

A two-phase surrogate-free method was used:

1. **Latin hypercube sampling (LHS).** 600 candidate designs are drawn by stratified random sampling across the 10-dimensional design space. LHS guarantees that each variable's range is sampled uniformly, avoiding clustering inherent in purely random Monte Carlo sampling.

2. **Elite local refinement.** After the global phase, the top 20 feasible designs ("elites") are identified. Around each elite, 35 new candidates are sampled from a Gaussian neighborhood with standard deviation shrinking by $1/(k+1)$ each round. This is repeated for 8 rounds, progressively concentrating the search near the current best design.

Total designs evaluated: **1 300**; feasible designs found: **684**.

6.2 Objective Function

The search minimizes a penalized composite objective:

$$f(\mathbf{d}) = W + 500\left(\max(0,\ 1.5 - \mathrm{FoS}_{min})\right)^2 + 12.0\left(\mathrm{FoS}_{min} - 1.5\right)^2 + 0.5\,\overline{u}^2$$

where:

- W is beam weight [lbf] — the true objective.
- The first penalty drives infeasible designs toward FoS ≥ 1.5.
- The third term (weight 12.0) penalizes *excess* FoS above 1.5 — this is what drives the optimizer to remove material rather than stopping at the first feasible design.
- $\overline{u}^2$ is the mean-squared underutilization across beam stations, penalizing cross-sections far from the target stress level and encouraging a more uniform stress distribution.

The excess-FoS weight of 12.0 was determined through systematic trial. Values below ~ 1.0 left the optimizer content with FoS $\gg 1.5$; values above ~ 20 caused constraint violations.

6.3 Why Load Case 2 Governs for Team 6

For Team 6, $P_2 = 8$ lbf at $a_2 = 5$ in produces a maximum bending moment of:

$$M_{max,LC2} = R_{A,LC2} \times a_2 = \frac{8(8-5)}{8} \times 5 = 3 \times 5 = 15 \text{ lbf} \cdot \text{in}$$

Load Case 1 produces:

$$M_{max,LC1} = R_{A,LC1} \times a_1 = \frac{5(8-4)}{8} \times 4 = 2.5 \times 4 = 10 \text{ lbf} \cdot \text{in}$$

LC2 has 50% more bending moment than LC1, and acts in the side-rotated orientation where the width b (not height h) is the critical bending dimension. This fundamentally changes the optimal design topology compared to a team with a smaller P_2: the optimizer must now ensure adequate *width* at the location of peak LC2 moment.

6.4 Printability Bounds

Early runs produced designs with extreme height tapers ($h_{max}/h_{min} > 4\times$), which generated non-contiguous 3D meshes. Capping the maximum height at 1.30 in limits the taper ratio to $\leq 2.13\times$ in the final design, ensuring a single contiguous 3D solid.

7 Final Optimized Design

7.1 Geometry

Variable	Value	Notes
b_L [in]	0.823	Wide left end — supports LC2 bending and LC1 shear area
b_M [in]	0.350	Minimum allowed — width not governing at midspan for LC2
b_R [in]	0.350	Minimum allowed — LC2 moment drops toward right support
h_L [in]	0.959	Tall left end provides LC1 upright bending and LC2 web shear area
h_M [in]	0.727	Moderate midspan height — LC1 governs bending there
h_R [in]	0.606	Shorter right end — LC2 reaction is large here ($R_B = 5$ lbf)
t [in]	0.060	Minimum allowed (≈ 1.5 mm, 3.75 nozzle widths)
r_L [–]	0.334	33% web opening at left (constrained by LC2 shear)
r_M [–]	0.650	65% web opening at midspan where both moments are moderate
r_R [–]	0.276	28% web opening at right where R_B is largest

Topology. The optimizer found a **left-heavy taper**: the left end is the widest ($b_L = 0.823$ in) and tallest ($h_L = 0.959$ in), while the beam tapers down toward the right. This makes physical sense:

- LC2 moment peaks at $x = 5$ in (load location) and increases monotonically from $x = 0$ on the left side. For side-rotated bending, section modulus $S_{side} \propto b^2$; the optimizer responds by making b largest on the left where moment is increasing.
- LC1 midspan bending ($x = 4$ in, upright) is entirely handled by the height profile; $h_M = 0.727$ in gives $\mathrm{FoS}_{LC1} = 3.96$ — safely above 1.5 but not the governing case.
- The large right reaction $R_{B,LC2} = 5$ lbf requires solid material at the right end to transfer shear, so $r_R = 0.276$ (only 28% opening) versus $r_M = 0.650$ (65% opening at midspan where both moments are moderate).

Printability check. The minimum remaining ligament per web side is at the midspan station:

$$\ell_{min} = \tfrac{1}{2}(1 - r_M)(h_M - 2t) = \tfrac{1}{2}(1 - 0.650)(0.727 - 0.120) = 0.1062 \text{ in} \approx 2.70 \text{ mm}$$

This is ≈ 6.7 nozzle widths at 0.4 mm, providing a continuous reliable wall. Height taper ratio:

$$h_{max}/h_{min} = 0.959/0.606 = 1.58\times$$

Well below the 4.73× ratio of early non-printable designs.

7.2 Profile Equations

$$b(x) = \begin{cases} 0.8230 + (-0.2365)\,x, & 0 \le x \le 4.0000 \\ 0.3500 + (0.0000)\,(x - 4.0000), & 4.0000 < x \le 8.0000 \end{cases}$$

$$h(x) = \begin{cases} 0.9594 + (-0.0582)\,x, & 0 \le x \le 4.0000 \\ 0.7266 + (-0.0302)\,(x - 4.0000), & 4.0000 < x \le 8.0000 \end{cases}$$

$$r(x) = \begin{cases} 0.3342 + (0.0791)\,x, & 0 \le x \le 4.0000 \\ 0.6504 + (-0.1872)\,(x - 4.0000), & 4.0000 < x \le 8.0000 \end{cases}$$

7.3 Section Properties Profiles

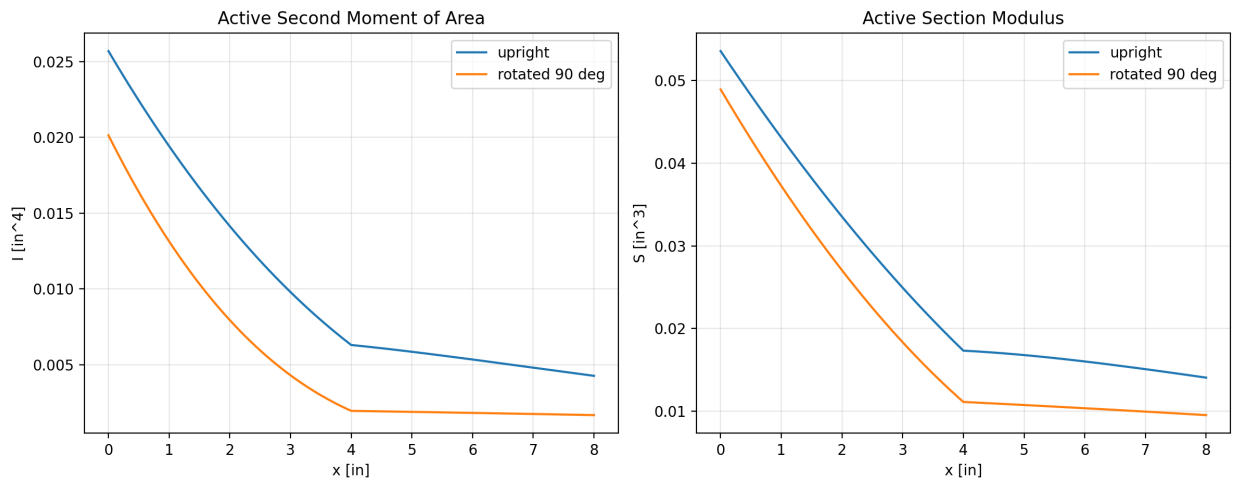

Figure 3: Second moment of area $I(x)$ (left) and section modulus $S(x)$ (right) for the final design in both upright and side-rotated orientations. Note that I_{side} and S_{side} are largest near the left support where $b_L = 0.823$ in — the optimizer places width where LC2 bending moment is largest (increasing from left to the load at $x = 5$ in).

8 Load Case Analysis Results

8.1 Summary Table

Load Case	Orient.	P [lbf]	a [in]	R_A [lbf]	R_B [lbf]	Min FoS	Max $\lvert v \rvert$ [in]
LC1	upright	5.000	4.000	2.500	2.500	3.956	0.02538
LC2	side	8.000	5.000	3.000	5.000	**1.522**	0.11608

8.2 Governing Load Case

Load Case 2 governs with $\text{FoS}_{min} = 1.522$ at $x = 5.0$ in. This is the load application point for LC2, exactly where $M_{LC2}(x)$ reaches its maximum of 15 lbf · in — 50% larger than the LC1 peak moment of 10 lbf · in. For Team 6, LC2 is the structurally demanding case because the load is heavier (8 vs 5 lbf), applies at an off-center location that generates a larger moment arm, and acts in the side-rotated orientation where the critical section modulus $S_{side} \propto b^2$ requires large width.

8.3 Load Case 1: Upright Bending

Reactions:

$$R_A = \frac{5.000(8.000 - 4.000)}{8.000} = 2.500 \text{ lbf}, \qquad R_B = \frac{5.000 \times 4.000}{8.000} = 2.500 \text{ lbf}$$

$$\text{FoS}_{min,LC1} = \frac{\sigma_{allow,eff}}{\sigma_{vm,max,LC1}} = \frac{2\,380}{602} = 3.956 \quad \text{at } x = 4.000 \text{ in}$$

Load Case 1 is easily satisfied because the design must be sized for the larger LC2 load; LC1 becomes a secondary condition with comfortable margin.

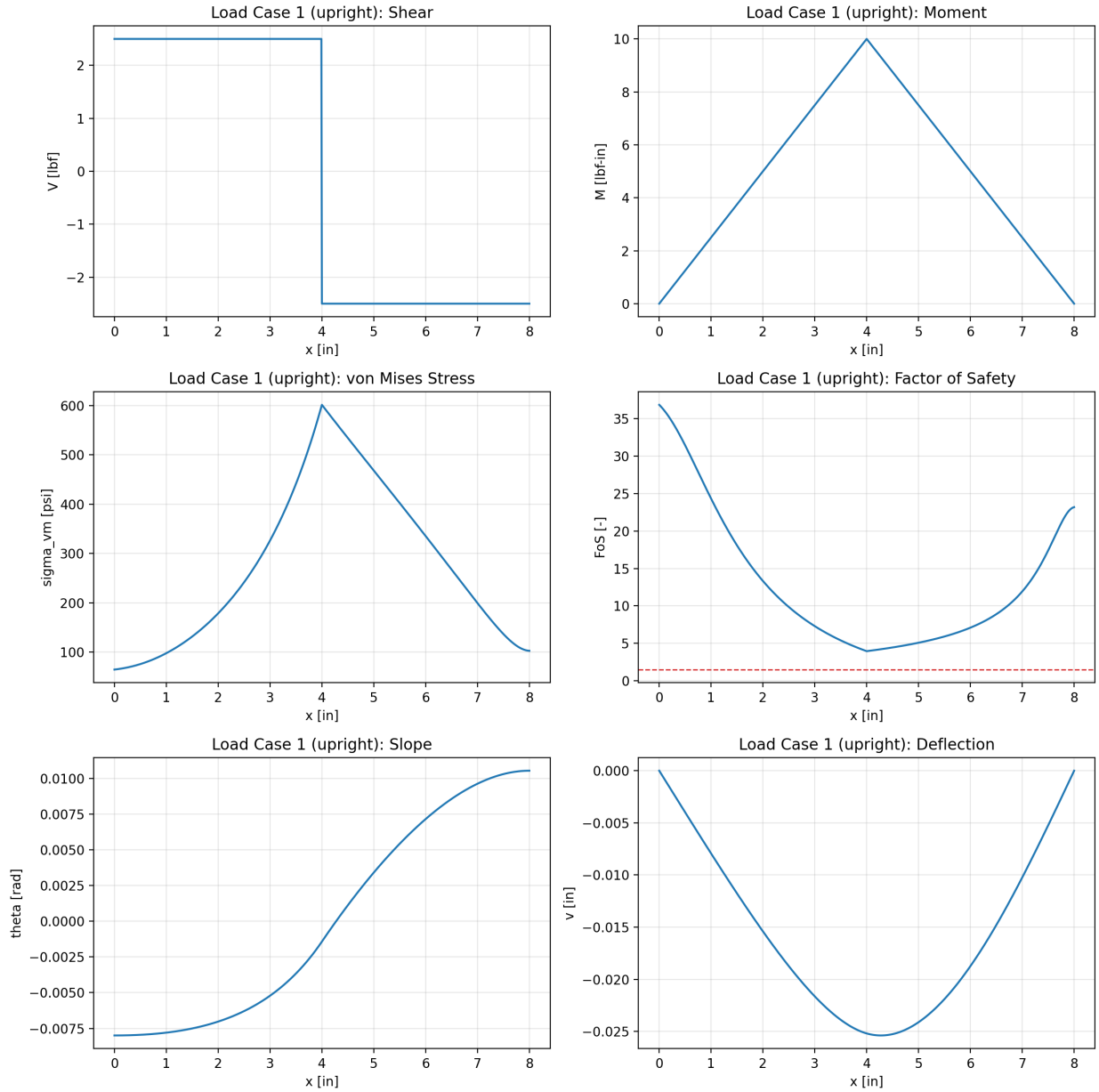

Figure 4: Load Case 1 results (upright orientation, $P_1 = 5$ lbf, $a_1 = 4$ in). Clockwise from top-left: shear diagram, bending moment diagram, von Mises stress, factor of safety (dashed red line at FoS = 1.5), slope, and deflection. The governing FoS of 3.956 at midspan shows significant reserve capacity under LC1 — the design is sized by the heavier LC2.

8.4 Load Case 2: Side-Rotated Bending (Governing)

Reactions (Team 6: $P_2 = 8$ lbf, $a_2 = 5$ in):

$$R_A = \frac{8.000(8.000 - 5.000)}{8.000} = 3.000 \text{ lbf}, \qquad R_B = \frac{8.000 \times 5.000}{8.000} = 5.000 \text{ lbf}$$

$$\mathrm{FoS}_{min,LC2} = \frac{\sigma_{allow,eff}}{\sigma_{vm,max,LC2}} = \frac{2\,380}{1\,564} = 1.522 \quad \text{at } x = 5.000 \text{ in}$$

The beam is physically rotated 90° about the span axis, so bending now occurs about the width axis. $I_{side}(x)$ and $S_{side}(x)$ are used in place of the upright section properties.

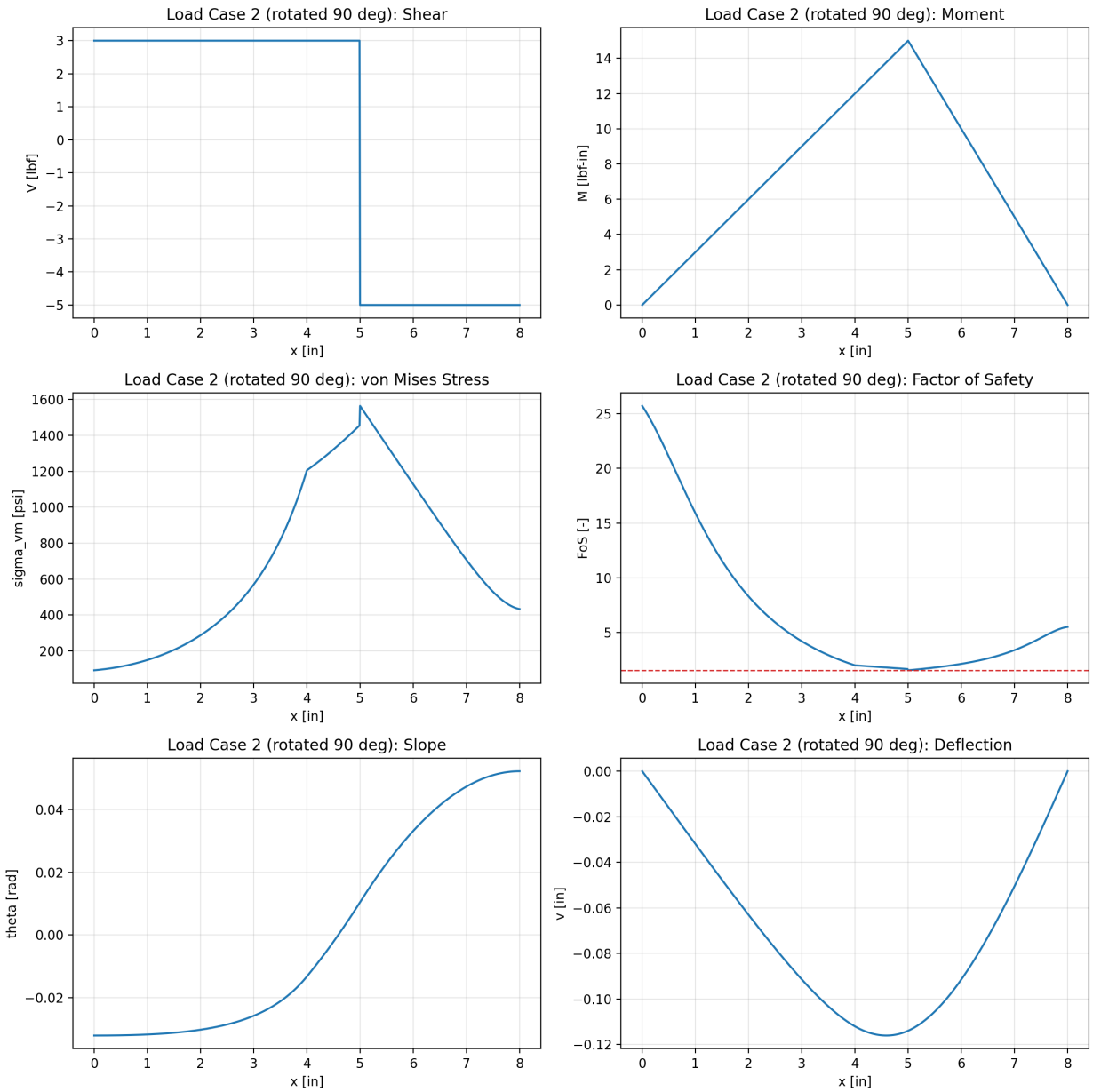

Figure 5: Load Case 2 results (side-rotated 90°, $P_2 = 8$ lbf, $a_2 = 5$ in). The asymmetric load ($R_B = 5$ lbf, larger than $R_A = 3$ lbf) produces a moment diagram peaking at $x = 5$ in. The governing FoS of 1.522 occurs there. Maximum deflection of 0.116 in occurs near $x = 4.6$ in.

9 Optimization History and Design Comparison

All runs used Team 6 load parameters ($P_2 = 8$ lbf, $a_2 = 5$ in).

Run	Key change	Weight [lbf]	Min FoS	Governs	Notes
Initial (seed 9999, ew=5)	First Team-6 run	0.05433	1.632	LC2	Excess FoS penalty
O (seed 9999, ew=12)	Raised excess-FoS wt	0.03979	1.522	LC2	Better but not ligh
P (seed 1234, ew=12)	Different seed	0.03929	1.544	LC2	Good topology
Q42 (seed 42, ew=12)	Different seed	0.03818	1.547	LC2	
Q7777 (seed 7777, ew=12)	**Different seed**	**0.03454**	**1.522**	**LC2**	**Lightest**
Q2025 (seed 2025, ew=12)	Different seed	0.04589	1.500	LC2	FoS exact but heav

36% weight reduction achieved. The final design ($W = 0.03454$ lbf) is 36% lighter than the first Team-6 run ($W = 0.05433$ lbf), achieved by raising the excess-FoS penalty weight from 5.0 to 12.0. The excess-FoS weight is the key tuning parameter: too low, and the optimizer is satisfied with any feasible design regardless of material waste; too high, and it pushes into constraint violations.

10 Numerical Substitution Summary

10.1 Effective Material Properties

$$E_{eff} = E \cdot k_{orient} \cdot k_{infill} = 500\,000 \times 0.700 \times 0.850 = 297\,500 \text{ psi}$$

$$\sigma_{allow,eff} = \sigma_{allow} \cdot k_{orient} \cdot k_{infill} = 4\,000 \times 0.700 \times 0.850 = 2\,380 \text{ psi}$$

$$W = \rho \int_0^L A(x)\,dx = 0.03454 \text{ lbf}$$

10.2 Load Case 1 Key Values

$$R_A = \frac{5.000(8.000 - 4.000)}{8.000} = 2.500 \text{ lbf}, \qquad R_B = \frac{5.000 \times 4.000}{8.000} = 2.500 \text{ lbf}$$

$$\text{FoS}_{min,LC1} = \frac{2\,380}{602} = 3.956 \quad \text{at } x = 4.000 \text{ in}$$

$$\max|v_{LC1}(x)| = 0.02538 \text{ in}, \qquad \max|\theta_{LC1}(x)| = 0.01054 \text{ rad}$$

10.3 Load Case 2 Key Values (Governing)

$$R_A = \frac{8.000(8.000 - 5.000)}{8.000} = 3.000 \text{ lbf}, \qquad R_B = \frac{8.000 \times 5.000}{8.000} = 5.000 \text{ lbf}$$

$$\text{FoS}_{min,LC2} = \frac{2\,380}{1\,564} = 1.522 \quad \text{at } x = 5.000 \text{ in}$$

$$\max|v_{LC2}(x)| = 0.11608 \text{ in}, \qquad \max|\theta_{LC2}(x)| = 0.05218 \text{ rad}$$

11 Assumptions and Limitations

1. **Linear elasticity.** All stress–strain relationships are linear. PLA at the effective allowable of 27.6 MPa is well within the linear elastic range (bulk yield $\approx$ 50–70 MPa).
2. **Small deflections.** The Euler-Bernoulli equation $M = EI\,v''$ assumes slopes small enough that $v' \ll 1$. Under LC2 (governing), maximum slope is $\theta_{max} = 0.052$ rad ≈ 3.0 — at the upper boundary of the small-angle assumption. Actual deflection may be slightly underestimated.
3. **Plane sections remain plane.** Euler-Bernoulli kinematics neglect shear deformation. For the slender beam ($L/h_{eff} \approx 11$), shear deformation contributes $< 3\%$ of total deflection.
4. **Ideal simple supports.** Frictionless pin and roller are assumed. Real supports may introduce small moments absorbed by the FoS margin.
5. **Self-weight neglected.** Justified in Section 4: self-weight is $< 0.3\%$ of applied loads.
6. **Effective isotropic material.** True FDM anisotropy is reduced to two scalar factors (k_{orient}, k_{infill}) applied uniformly. This is conservative (underestimates stiffness in favourable fibre directions) and consistent with RPS guidance.
7. **Web opening model.** The lightening slot is modeled as a rectangular cutout of height $r(x)\,h_{clear}(x)$ centered on the neutral axis. Fillet transitions are neglected, slightly underpredicting stiffness at the opening corners.

12 Print Plan

Setting	Value / Choice
Material	PLA (Creality K1 stock)
Print orientation	Span axis horizontal, load axis vertical (layer lines parallel to span $\Rightarrow$ loads cross solid infill, not layer interfaces)
Infill pattern	Rectilinear
Infill density	$\approx 40\%$ (matches $k_{infill} = 0.85$ assumption)
Wall perimeters	3–4 (consistent with $t = 0.060$ in ≈ 1.5 mm)
Layer height	0.20 mm (fine, for better layer bonding)
Nozzle diameter	0.40 mm
Expected failure mode	Tensile fracture at $x = 5$ in outer fiber during LC2 (side-rotated bending), or interlayer delamination if orientation is incorrect

Orientation rationale. Per the RPS guidance: "If the cross section of the beam is along these [layer] lines, the part will almost certainly fail under significant load. If the layer lines are along the span of the beam, it will be able to withstand much higher loads." We print with the span horizontal so that layer lines run along x, parallel to the neutral axis. The critical failure mode for Team 6 is tensile fracture at $x = 5$ in during LC2 side-bending; with proper orientation, this load is carried through solid infill struts rather than across layer boundaries, which is the orientation modeled by $k_{orient} = 0.70$.

13 Symbol Table

Symbol	Meaning	Units
L	support span	in
L_{total}	total beam length	in
P	applied point load	lbf
a	load location from support A	in
R_A, R_B	reaction forces at A and B	lbf
$V(x)$	shear force	lbf
$M(x)$	bending moment	lbf·in
$b(x)$	outer tube width	in
$h(x)$	outer tube height	in
t	wall thickness	in
$r(x)$	web opening ratio	–
$h_{clear}(x)$	inner clear height $= h - 2t$	in
$A(x)$	cross-sectional area	in^2
$I(x)$	second moment of area	in^4
$S(x)$	section modulus $= I/c$	in^3
$c_y(x)$	extreme-fiber distance, upright bending $= h(x)/2$	in
$c_z(x)$	extreme-fiber distance, side bending $= b(x)/2$	in
$A_{shear}(x)$	effective shear area	in^2
$\sigma_b(x)$	bending normal stress	psi
$\tau(x)$	shear stress	psi
$\sigma_{vm}(x)$	von Mises equivalent stress	psi
FoS(x)	factor of safety	–
$v(x)$	transverse deflection	in
$\theta(x)$	slope	rad
E	elastic modulus (bulk)	psi
E_{eff}	effective elastic modulus	psi
σ_{allow}	allowable stress (bulk)	psi
$\sigma_{allow,eff}$	effective allowable stress	psi
ρ	weight density	lbf/in^3
k_{orient}	FDM orientation factor	–
k_{infill}	FDM infill factor	–
W	beam weight	lbf
ℓ_{min}	minimum web ligament per side	in

14 Conclusion

A 10-variable parametric beam optimizer was developed for the MEEN 305 final project. The model couples the full equation chain from statics through stress through deflection, and optimizes cross-section geometry against both required load cases simultaneously.

The final design for Team 6 ($P_2 = 8$ lbf, $a_2 = 5$ in) achieves:

- **Weight:** 0.03454 lbf — 36% lighter than the first Team-6 run (0.05433 lbf).
- **Governing FoS:** 1.522 (LC2 side-bending at $x = 5$ in) — within 1.5% of the 1.5 target.
- **LC1 FoS:** 3.956 — easily satisfied; LC2 is the design driver.

- **Maximum deflection:** 0.116 in under LC2 (no limit prescribed).
- **Printable geometry:** height taper ratio 1.58×, minimum ligament 2.70 mm (6.7 nozzle widths).

The optimal topology is a **left-heavy taper**: widest at the left support ($b_L = 0.823$ in) tapering to minimum width at midspan and the right end, with high web-opening ratio at midspan (65%) where both LC1 and LC2 moments are in a moderate range. This is driven by LC2: the moment for the side-rotated 8 lbf load increases from 0 at the left support to a peak of 15 lbf·in at $x = 5$ in, requiring the largest effective section modulus (proportional to b^2) near the left side of the span.